Dr. HUANG Qirui

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Hiring Team, AI Infrastructure Lead (AI Inference Platform)
Gateway Search, Singapore

Dear Hiring Manager,

I am writing to apply for the AI Infrastructure Lead (AI Inference Platform) role in Singapore. I am Head of Huawei's Central Research Institute (2012 Labs) in Singapore, and over 15 years my work has moved up the stack an inference platform is built on: data-center networks and cloud infrastructure, then the systems that train and serve large models, and now the memory and caching layers that decide what serving costs.

On the serving side, I led development of the training and inference engine for Huawei's Pangu models, which runs 10x faster than Microsoft DeepSpeed, and I direct research on large GPU clusters for LLM and multimodal training and inference, from parallelism and mixed precision to long-context serving. In vLLM, I built a unified KV and recurrent-state prefix cache for hybrid LLMs whose restored outputs are bitwise-identical to recomputation. I now lead an agent memory systems program that covers shared memory across agents, KV and state caching, and moving knowledge between text, KV cache and LoRA across GPU, DRAM, SSD and HDD tiers. These are the trade-offs a token factory lives with every day: what to keep in GPU memory, what to offload, and when it is cheaper to recompute.

I stay hands-on with infrastructure. I built a Kubernetes-based containerized GPU training cluster for our 6.3B hybrid Mamba-Transformer MoE model. At SUTD I designed, built and operated a research testbed of tens of racks and a hybrid cloud of thousands of VMs; at A*STAR I worked on optical switching for data centers and supercomputers; and for my PhD I wrote an open-source optical packet switching simulator in C++. I work in Python, C/C++ and Triton.

I have also built teams from zero. I founded and built Huawei's Singapore lab from the ground up (the Singapore region has 40 staff), later handing it over to a Chief Expert, and I now lead a five-person core team. I lead Singapore's R&D within a globally distributed institute with teams in the UK, Canada, Switzerland and Japan, and I hire and mentor researchers. Earlier, at SUTD, I led a 20-person team on the CISDeM project and delivered a MINDEF-sponsored research programme worth SGD 4.7M.

The Token Factory platform is the kind of work I want to do next: setting the architecture from the first design review, turning a global GPU fleet into reliable and cost-efficient tokens through scheduling, caching, serving and observability, and building the Singapore team that runs it.

Thank you for your consideration. I would be glad to discuss how I can contribute.

Sincerely,

Dr. HUANG Qirui